

KOK HUA NG

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EDUCATION

University of Washington

Seattle, WA

B.S. in Computer Science and Software Engineering | GPA: 3.85

Expected June 2027

- Relevant Coursework: **Data Structures & Algorithms, Software Engineering, Analysis & Design, Information**

Assurance & Cybersecurity, Hardware & Computer Organization

Bellevue College

Bellevue, WA

Associate in Arts and Sciences | GPA: 3.98

July 2023 – June 2025

EXPERIENCE

Pullscription

Los Angeles, CA

Software Engineering Intern

June 2024 – Sep 2024

- Built **10+ reusable SwiftUI components**—custom buttons and collection views—ensuring consistent, **responsive UI** across all screen sizes and orientations.
- Integrated **REST API** endpoints to retrieve and display dynamic content on the **iOS Explore** page, including **secure store authentication**.
- Developed and tested **backend** endpoints supporting **data retrieval**, store communication, and robust **error-handling** workflows.

University of Washington — Community Data Science Collective

Seattle, WA

Undergraduate Research Assistant

June 2026 – Present

- Conducting **mixed-methods research** on **AI-generated cybersecurity risks** in open-source software, investigating how fabricated **LLM bug reports** cause **maintainer burnout** and disrupt **triage workflows**.
- Built a **dataset of real bug reports** using the **GitHub Advisory Database** and **GitHub API**, sampling **7,780+ validated repositories** split into pre/post-**ChatGPT** cohorts for **comparative analysis**.
- Simulated **adversarial AI-generated bug reports** using multiple **LLMs** to evaluate accuracy of **AI detection tools**; findings contributed to a **research poster** presented at UW.

Xtatic

Bothell, WA

Math Instructor & Robotics Club Teaching Assistant

Sep 2025 – Present

- Instructed **20+ students** in **algorithmic problem-solving** and **discrete mathematics**, communicating complex concepts clearly and adapting to diverse skill levels.

Bellevue College

Bellevue, WA

Teaching Assistant

Nov 2024 – June 2025

- Guided students through **calculus, algebra, and statistics**, reinforcing **analytical problem-solving** and adapting **instructional strategies** to diverse learning styles.

PROJECTS

AI Voice Assistant | Python, LiveKit Agents, OpenAI API, Silero VAD, WebSocket

2025

- Built a **real-time conversational AI** pipeline using **LiveKit Agents** chaining **OpenAI STT, LLM, and TTS** over **WebSocket** for low-latency end-to-end voice interaction.
- Integrated **Silero VAD** for **voice activity detection** and designed **async function calling** to get/set temperatures across **5 smart-home zones** via natural speech commands.

Personal Portfolio Website | HTML5, CSS3, JavaScript, Netlify, Responsive Design

2025

- Built a **fully responsive** portfolio with **HTML5, CSS3, and vanilla JavaScript**, handling **desktop, tablet, and mobile** breakpoints via a dedicated **mediaqueries.css** file.
- Implemented **smooth-scroll navigation**, a **hamburger menu** for mobile UX, and **CSS animations**; deployed live at kok-hua-ng.netlify.app via **Netlify**.

File Management Automation | Python, Watchdog, OS, Logging

2025

- Developed a **real-time file monitoring** tool using the **Watchdog** library to watch a directory and **auto-classify** files (images, videos, documents, audio) into typed destination folders.
- Built **duplicate filename handling, permission-error resilience, and timestamped logging**; added **smart audio routing** by file size to separate SFX from music libraries.

Open Source Contributor | Java, TypeScript, i18n, Git — OpenHospital GUI & UI

2026

- Merged PR #2202 into **openhospital-gui**: fixed an **XMPP i18n key typo** and standardized **canonical key naming** across all locale bundles, eliminating duplicate keys and preventing future **localization drift**.
- Opened PR #771 in **openhospital-ui**: audited **12 locale bundles** via **static grep analysis** across the Java codebase, removing **44 confirmed orphaned i18n keys (528 deletions)** to reduce **technical debt**.

SKILLS

Languages: Python, JavaScript, C++, SQL, Java, Swift | **Frameworks:** React, Node.js, SwiftUI, LiveKit

Tools: AWS, GitHub, Xcode, VS Code | **Concepts:** REST APIs, OOP, CI/CD, Agile